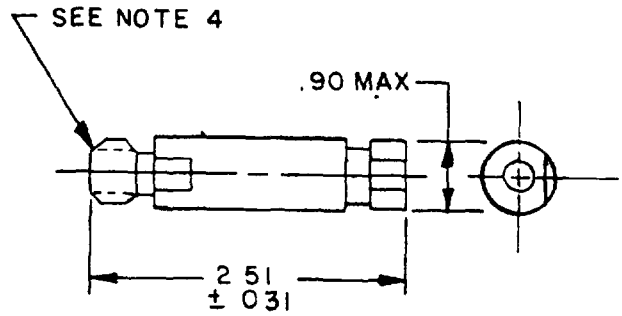
# MILITARY SPECIFICATION SHEET

## CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SERIES N (SOCKET CONTACT, SEMIRIGID CABLE, CLASS 2) (FOR USE WITH MIL-C-28830/3)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connector described herein shall consist of this specification and the latest issue of MIL-C-39012.



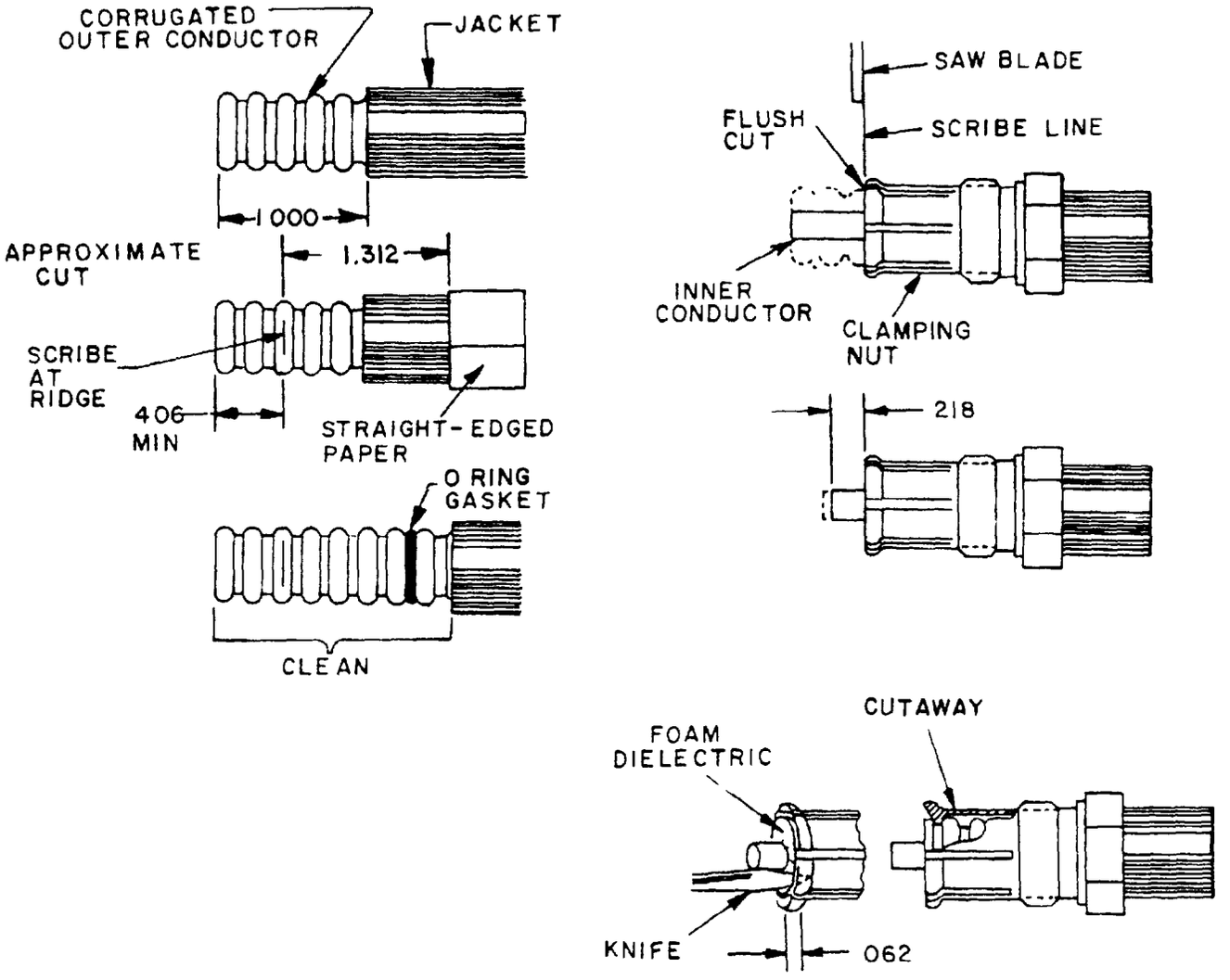

| Inches | mm |
|---|---|
| 031 | 0 79 |
| 90 | 22 86 |
| 2.51 | 63 75 |

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only
3 All undimensioned pictorial configurations are for reference purposes only.
4 Series N, socket contact interface per MIL-STD-348

FIGURE 1 General configuration.



AMSC N/A FSC 5935
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| Inches | mm |
|---|---|
| 062 | 1 57 |
| 218 | 5 54 |
| 406 | 10 31 |
| 1 000 | 25 40 |
| 1 312 | 33 32 |

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Cable outer conductor to be cut at crest only

FIGURE 2. Cable stripping dimensions for connectors.



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ENGINEERING DATA·

Nominal impedance· 50 ohms.

Frequency range 0 to 8.1 GHz.

Voltage rating· 500 volts.

Temperature range: -40°C to +100°C.

Peak RF power: 5.0 kW.

REQUIREMENTS

Dimensions and configuration. See figure 1.

Force to engage and disengage:

Longitudinal force Not applicable.

Torque· 6 inch pounds, maximum.

Coupling proof torque: 15 inch pounds, minimum.

Inspection conditions: 6 - 10 inch pounds.

Mating characteristics. See MIL-STD-348 for dimensions.

Hermetical seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance· Method 302, test condition B, MIL-STD-202.

5,000 megohms, minimum.

Center contact retention 6 pounds, minimum.

Radial torque. Not applicable.

Corrosion (salt spray): Method 1001, test condition C, in accordance with MIL-STD-1344.

Voltage standing wave ratio (VSWR) per connector: 0.5 to 8.0 GHz-1.2.1.

Swept frequency VSWR test setup:

Item 6· VSWR shall be less than 1.008 +.002 f (f in GHz).

Item 16 VSWR shall be less than 1.008 +.002 f (f in GHz).

Second step of VSWR checkout procedure· VSWR shall be less than 1.012 +.005 f (f in GHz).

Group B inspection 1.024 +.007.

Qualification and group C inspection· Shall not exceed 1.08.

Connector durability A minimum of 500 cycles at 12 cycles per minute, maximum.

Contact resistance In milliohms, maximum.

| | Inital | After environment |
|---|---|---|
| Inner contact: | 2.0 | 2.0 |
| Outer contact· | 0.3 | 0.3 |



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Dielectric withstanding voltage at sea level: 2,000 volts, method 301 of MIL-STD-202.

Vibration, high frequency Method 204, test condition D, MIL-STD-202. Cable clamps shall be mounted on the vibration table 6 inches minimum from the mating face of the connector pair.

Shock Method 213, test condition 1, MIL-STD-202.

Thermal shock Method 107 of MIL-STD-202 test condition B, except test high temperature shall be +115°C.

Moisture resistance Method 106 of MIL-STD-202.

No measurements at high humidity. Insulation resistance shall be of at least 200 megohms within 5 minutes after removal from humidity.

Corona level. Not applicable.

RF high potential withstanding voltage: 1,500 volts rms.

Frequency: 5 - 7 MHz.

Leakage current: Not applicable.

Cable retention force 200 pounds, minimum.

Torque 55 inch ounces.

Coupling mechanism retention force 100 pounds, minimum.

RF leakage: -90 dB tested at a frequency of 2 - 3 GHz.

RF insertion loss dB maximum - 0.10 dB maximum at 4.0 GHz.

Part number M39012/118-0001

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians
Army - CR
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - SH
Air Force - 11, 15, 99

User activities
Army - AM, AT
Navy - AS, MC, OS
Air Force - 90

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5935-3568-01)

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